IskolE

The School Management System

SECOND YEAR GROUP PROJECT REPORT

SCS2202 – Group Project

CS Group: 38

Group Member Details

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Contents

1 Introduction **6**

1.1 Domain Description . . . 6

1.2 Current System and Limitations . . . 6

1.3 Goals & Objectives . . . 7

1.4 Assumptions . . . 7

2 Feasibility Study **8**

2.1 Technical Feasibility . . . 8

2.1.1 Front-End Technologies . . . 8

2.1.2 Back-End Technologies . . . 8

2.1.3 Environmental Technologies . . . 8

2.1.4 Database Management . . . 8

2.1.5 Version Control and Collaboration . . . 9

2.2 Schedule Feasibility . . . 9

2.2.1 Project Timeline . . . 9

2.2.2 Deliverables . . . 9

2.2.3 Resource Availability . . . 9

2.2.4 Tools Used for Planning . . . 10

2.3 Resource Feasibility . . . 10

2.3.1 Human Resources . . . 10

2.3.2 Technical Resources . . . 10

2.3.3 Knowledge and Skills . . . 10

2.4 Operational Feasibility . . . 10

2.5 Economic Feasibility . . . 11

2.6 Legal & Ethical Feasibility . . . 11

3 Requirements Specification **13**

3.1 Stakeholders / Actors . . . 13

3.2 Functional Requirements . . . 13
3.2.1 Parent . . . 13
3.2.2 Administrator . . . 15
3.2.3 Students . . . 16
3.2.4 Management Panel . . . 16
3.2.5 Teacher . . . 18
3.3 Non-functional Requirements . . . 19
3.4 In-Scope vs Out-of-Scope . . . 20
3.4.1 In-Scope . . . 20
3.4.2 Out-of-Scope . . . 21
3.5 Constraints & Limitations . . . 21

4 System Architecture 23
4.1 Components & Responsibilities . . . 23
4.1.1 Model (M) . . . 23
4.1.2 View (V) . . . 23
4.1.3 Controller (C) . . . 24
4.2 Key Modules & Interactions . . . 24

5 System Design 26
5.1 Use Case Diagram . . . 26
5.2 ER Diagram . . . 26
5.3 Class Diagram . . . 26
5.4 System Architecture Diagram . . . 26
5.5 Activity Diagrams . . . 34

6 Completeness of the Project 40
6.1 Functionalities Completed . . . 40
6.1.1 R K K Jinendra – Module Owner . . . 40
6.1.2 Seniru D S Senaweera – Module Owner . . . 40
6.1.3 S. K. Thasindu Ramsitha – Module Owner . . . 40
6.1.4 R. S. R. G. A. A. Ananda – Module Owner . . . 41
6.2 Functionalities Yet to Complete . . . 41

7 Individual Contribution Analysis **42**

8 Testing **44**

8.1 Testing Types 44

8.2 Test Case Summary 45

9 Details of Project Supervisor and Co-Supervisor **46**

10 Declaration **47**

List of Figures

5.1 Administrator Use Case Diagram 27
5.2 Management Panel Use Case Diagram 28
5.3 Parent Use Case Diagram 29
5.4 Student Use Case Diagram 30
5.5 Teacher Use Case Diagram 31
5.6 ER Diagram 32
5.7 Class Diagram 33
5.8 Admin Workflow 34
5.9 Manager Workflow 35
5.10 Parent Workflow 36
5.11 Student Workflow 37
5.12 Teacher Workflow 38
5.13 User Login and Password Reset Workflow 39

7.1 Individual Contribution of Team Members 42

List of Tables

4.1 Key System Modules and Their Interactions 24

7.1 Individual Contribution Summary 43

8.1 Testing Summary 45

Chapter 1

Introduction

IskolE is a web-based School Management System (SMS) designed to streamline academic, administrative, and communication workflows for Sri Lankan schools. It centralises core activities such as marks/grade management, attendance & leave, timetabling, announcements, learning materials, and performance analytics while enforcing role-based access for Administrators, Management, Teachers, Parents, and Students.

This report consolidates the proposal into a complete software engineering document covering feasibility, requirements, architecture (using the MVC pattern), system design, implementation completeness, testing, and individual contributions.

1.1 Domain Description

The domain is school operations: academic record keeping; parent–teacher–student communication; attendance and leave handling; class, exam, and annual academic scheduling; and dissemination of learning materials and announcements. The system's stakeholders interact via a secure web application with role-scoped permissions.

1.2 Current System and Limitations

Many schools still rely on paper logs or ad-hoc spreadsheets and verbal/noticeboard communication. This leads to:

- Data entry errors
- Poor transparency of performance and attendance
- Delayed reporting

- Manual compilation overhead for administrators
- Fragmented communication channels

1.3 Goals & Objectives

- Improve communication via structured announcements and dashboards.
- Provide real-time access to marks, ranks, and progress analytics for authorised users.
- Centralise documents and learning materials.
- Automate timetables and event scheduling with reminders.
- Support behaviour/remarks logs and comprehensive student profiles.
- Enforce secure, role-based access and privacy.
- Reduce administrative overhead through automation and reporting.

1.4 Assumptions

- Users have internet access and valid emails.
- One role per account; unique email; role-scoped data access.
- The school must provide the foundation data, and the system builds everything else on top of it.

Chapter 2

Feasibility Study

2.1 Technical Feasibility

In this section we measure how feasible our solution is in a real-world scenario using selected technologies and methods.

2.1.1 Front-End Technologies

- **HTML, CSS, and JavaScript:** These technologies are used to create a responsive and interactive user interface.

2.1.2 Back-End Technologies

- **PHP:** PHP is used for handling server-side logic, managing user sessions, server-side requests, and interacting with the MySQL database.

2.1.3 Environmental Technologies

- **Docker:** Docker is used for containerisation and deployment of the application. This allows for easy deployment, scalability, and consistency across development and production environments.

2.1.4 Database Management

- **MySQL:** MySQL is used to manage databases; SQL queries execute via PHP from the backend.

2.1.5 Version Control and Collaboration

- **Git/GitHub:** The Git version control system and GitHub are used to facilitate collaborative contributions from every group member.
- **Jira:** Jira is used for task management and team collaboration, allowing efficient scheduling and development planning.

Most of the technologies mentioned above are freely available or open source. The team gained further knowledge and became more familiar with these technologies during the development phase. The free version of Jira was used for task management and team collaboration.

2.2 Schedule Feasibility

Schedule feasibility focuses on whether the project can be successfully completed within the given timeframe, using the available resources and technologies.

2.2.1 Project Timeline

This project spans a full academic year, with development distributed across the first and second semesters.

2.2.2 Deliverables

An interim presentation was delivered after the first semester examinations, showcasing all finalised and implemented user interfaces of the system. The complete system was developed and finalised for submission after the second semester examinations.

2.2.3 Resource Availability

All team members worked on this project alongside their regular academic responsibilities during both semesters. Time was planned to ensure consistent progress while balancing academic workload.

2.2.4 Tools Used for Planning

Jira was used to schedule tasks, assign responsibilities, and track progress throughout the project lifecycle, helping the team stay on track and meet all major deadlines.

2.3 Resource Feasibility

2.3.1 Human Resources

The project team consists of four dedicated members who worked collaboratively throughout the first and second semesters. Each member contributed their time and effort alongside academic responsibilities, ensuring steady progress during the academic year.

2.3.2 Technical Resources

Development and testing were carried out using the team's personal computers, which are adequately equipped for the project's needs.

2.3.3 Knowledge and Skills

The team gained foundational knowledge of relevant technologies during the first year of study and continued to expand understanding and practical skills as the project progressed, ensuring readiness for implementation and deployment phases.

2.4 Operational Feasibility

Focuses on efficiently managing resources, optimising workflows, mitigating risks, and maintaining accurate documentation to streamline project execution.

- **Resource Allocation:** Efficiently managing resources such as time, budget, and equipment.
- **Workflow Management:** Streamlining processes to maximise productivity.
- **Risk Management:** Identifying and reducing risks that could impact project success.

- **Documentation and Reporting:** Ensuring accurate and timely documentation of project progress and outcomes.

2.5 Economic Feasibility

Manages project costs within budget constraints, ensures financial sustainability and return on investment, and conducts cost-benefit analyses to justify project expenditures.

- **Cost Management:** Controlling project expenses within planned budgets.
- **Financial Sustainability:** Ensuring long-term financial viability and funding for the project.
- **Return on Investment:** Estimating the benefits and value generated by the project.

2.6 Legal & Ethical Feasibility

This section ensures that the project strictly adheres to school policies, ethical guidelines, and relevant legal regulations, maintaining integrity throughout the project lifecycle by addressing key areas:

- **Compliance:** The project complies with all applicable school policies, ethical standards, and legal requirements related to software development and data management.
- **Intellectual Property:** Managing ownership and rights related to project outcomes and innovations.
- **Data Protection:** Sensitive data collected or generated during the project, such as student information and login credentials, is securely handled to protect privacy and prevent unauthorised access.
- **Research Ethics:** Upholding ethical standards in data collection, analysis, and reporting.
- **Conflicts of Interest:** Managing conflicts of interest that may arise among project stakeholders.

- **Transparency:** Maintaining transparency in decision-making processes and outcomes.

Chapter 3

Requirements Specification

3.1 Stakeholders / Actors

The system identifies the following primary stakeholders:

1. Administrator
2. Management Panel
3. Teacher
4. Parent
5. Student

3.2 Functional Requirements

3.2.1 Parent

1. Exam and Report Management

- The system allows teachers to input student exam marks after each exam.
- Upon mark entry, the system automatically:
  - Calculates total marks.
  - Calculates average marks.
  - Determines student rank.
- The system generates a comprehensive student report.

- Only authorised users (teachers, respective students, their parents, and management) can view the student reports. Unauthorised users are restricted from accessing these reports.

2. Attendance and Absence Management

- Parents can submit pre-absence requests to inform teachers in advance about a child's absence.
- The submitted absence requests are visible to the relevant class or subject teacher.
- Parents can view their child's complete attendance history, including present, absent, and excused days.

3. Announcement Viewing

- Admin, Teachers and the management panel can publish announcements through the system.
- Users can view these announcements in the announcement preview panel.
- Announcements may include exam details, school events, holiday notices, and other important messages.

4. Subject Teacher Information

- Parents can view a list of teachers responsible for each subject their child studies.
- The system displays relevant contact information (e.g., name, email, phone) for each subject teacher.

5. User Authentication

- Parents can log into the system using initial credentials.
- These credentials are generated and distributed by the management panel.
- Unauthorised access is restricted through role-based access control.

6. Student Behaviour Records

- Teachers can log behavioural notes and observations about students into the system.
- Parents can view these behaviour records via their portal.

3.2.2 Administrator

1. **CRUD – Management Staff**
   - Allows the admin to Create, Read, Update, and Delete records of the principal, vice principal, and other management staff.
   - Admin logs into the system, navigates to the management panel, and performs the desired CRUD operations on staff records.
   - Management staff records are updated in the system.
2. **CRUD – Teachers**
   - Enables the admin to manage teacher profiles by performing CRUD operations.
   - Admin logs in and accesses the teacher management section.
   - Performs Create, Read, Update, or Delete operations as needed.
   - The teacher database is updated accordingly.
3. **CRUD – Students**
   - Allows the admin or authorised class teacher to manage student records.
   - Authorised user selects the student section and performs CRUD operations (add, view, update, delete).
   - Student information is updated and maintained in the system database.
4. **CRUD – Parents**
   - Enables the admin or teacher to manage parent/guardian information linked to students.
   - Authorised user accesses the parent section and executes Create, Read, Update, or Delete actions as needed.
   - Parent data is accurately stored and kept up to date in the system.
5. **Grade, Class, and Subject Management**
   - Administrator can create and manage grades, classes, and subjects within the system.

- Administrator can assign teachers to respective classes.

3.2.3 Students

1. **View Marks:** The system allows students to view their subject-wise marks, final examination marks.
2. **View Timetable:** The system allows students to view their daily and weekly timetables with real-time updates for schedule changes.
3. **Check Academic Progress:** The system enables students to view their academic progress.
4. **View Announcements:** The system allows students to view announcements posted by school administration and teachers.
5. **View Exam Time Table:** Student is able to view exam timetable that was uploaded by Management pannel or admin
6. **View Absent Reason:** The system allows parents/students to view the status of their submitted absence reasons.
7. **Login:** The system requires valid login credentials (email and password) to access the portal, authenticating user credentials against the database.
8. **View Result Report:** The system allows students to view detailed result reports including grades and rank.
9. **View Attendance:** The system allows students to view their attendance records with calculated attendance percentage.

3.2.4 Management Panel

1. User Login

- **User Authentication:** The system allows management panel users to log in using a valid email and password.
- **Credential Verification:** The system validates the entered credentials against stored user data in the database.

- **Access Control:** The system grants access to the management panel dashboard only if the credentials are correct.
- **Error Handling for Invalid Login:** The system displays an appropriate error message if the credentials are invalid (e.g., "Invalid email or password").

2. CRUD – Student Records

- The management panel user can search for a specific student and edit their records, including personal details, grades.
- The user initiates a search by entering identifying information (student ID or name).
- The system displays a list of matching student records; the user selects a student and modifies preferred fields.
- The system validates the input data (e.g., data type, required fields).
- The student's record is updated in the database with the validated changes, and the user is notified of the successful update.

3. CRUD – Parent Records

- The system retrieves the parent's contact details and other relevant information from the database and displays it clearly.

4. CRUD – Staff Records

- A management panel user logs into the system, accesses staff records, reviews details, and updates staff information such as name, role, contact info, and employment history.
- The system validates the input data for correctness and completeness. Upon successful validation, the system updates the staff record in the database and confirms the update.

5. Broadcast Announcements

- The system allows users with Management Panel or Administrator roles to create announcements with a title (mandatory), message content (mandatory), and target audience selection (mandatory).

- Input validation is performed before saving. The announcement is broadcast to all users specified in the target audience across the platform.
- The newly created announcement is displayed in the announcement module list for future reference.

6. **Monitor Attendance**
   - The user logs into the system, opens the attendance module, selects a teacher from the list, and the system retrieves and displays detailed attendance records showing dates and attendance status for review.
7. **Leave Requests**
   - The system displays all leave requests submitted by teachers with details including requester name, duration (from/to dates), and current status (Pending, Approved, Rejected).
   - The user reviews requests for potential further action.
8. **View Performance Trends**
   - User navigates to the academic performance module, selects a student or group of students, and the system retrieves academic records including grades, exam results, and progress reports for analysis and monitoring.

3.2.5 Teacher

1. **Teacher Authentication:** The system provides a login interface for teachers and verifies teacher credentials (email and password), granting access to authorised features upon successful login.
2. **View Student Reports:** The system allows teachers to select a student from their class list and displays student reports including academic marks, attendance records, and teacher remarks.
3. **Post Announcements:** The system allows teachers to post announcements to students and/or parents via a submission module.
4. **Submit Leave Requests:** The system provides a leave application form for teach-

ers to submit leave requests to management.

5. **Track Leave Request Status:** The system allows teachers to check the approval status of their leave requests (pending, approved, or rejected).
6. **Manage Announcements:** The system allows teachers to edit or delete previously posted announcements.
7. **Add Student Remarks:** The system allows teachers to select a student and add remarks to their record.
8. **Upload Learning Materials for Absent Students:** The system allows teachers to select students marked absent and upload worksheets, notes, or learning content.
9. **Manage Student Marks:** The system allows teachers to add or update student marks for assessments with validation before saving.

3.3 Non-functional Requirements

- **Security:**
  - Password hashing to protect user credentials.
  - Role-Based Access Control (RBAC) to restrict access based on user roles and permissions.
  - Session timeouts to automatically log out inactive users and prevent unauthorised access.
- **Reliability:**
  - The system ensures high availability and minimal downtime during normal operations.
  - Data consistency is maintained across all modules through transactional integrity.
  - Error handling mechanisms are implemented to gracefully manage unexpected failures.
  - Backup and recovery mechanisms are supported to prevent data loss.

- **Performance:**
  - Typical page load times under 2 seconds on standard broadband connections.
- **Usability:**
  - Intuitive and easy-to-use interface for all user roles.
- **Maintainability:**
  - Model-View-Controller (MVC) architectural pattern for clear code separation.
  - Readable and well-documented code with unit and integration tests.
  - Continuous Integration (CI) process to automate testing and improve development efficiency.
- **Portability:**
  - Deployable on a standard LAMP stack (Linux, Apache, MySQL, PHP).
  - Containerisation using Docker for enhanced portability across environments.

3.4 In-Scope vs Out-of-Scope

3.4.1 In-Scope

The following features and modules are included within the scope of the system development:

- **Academic Records Management:** Handling student profiles, grades, and academic history.
- **Attendance and Leave Management:** Recording student and staff attendance, and managing leave requests and approvals.
- **Announcements:** Providing a centralised platform for school-wide communications and updates.
- **Learning Materials:** Uploading and accessing course content, assignments, and educational resources.

- **Role-Based Access Control (RBAC):** Ensuring secure access based on user roles such as administrators, teachers, students, and parents.
- **Reporting and Analytics:** Generating reports on attendance and performance.

3.4.2 Out-of-Scope

The following functionalities are not included in the first phase of development but may be considered for future releases:

- Payroll and Finance Management
- Library and Inventory Systems
- Transport Management
- Health and Medical Records
- Non-Academic Staff Modules

3.5 Constraints & Limitations

- Each user is allowed to maintain only one account, and each account is assigned a single role (e.g., student, teacher, parent, administrator, or management panel).
- All users are required to possess a valid and unique email address for account creation, verification, and communication.
- The server and database are expected to remain available during normal operations, with minimal downtime.
- The fingerprint attendance module is a simulated feature for demonstration purposes. It is assumed that the fingerprint input correctly maps to the corresponding UserID.
- Phone number is used as the initial default password for new accounts.
- NIC format: Either 12-digit numeric.
- One class per student: Students belong to exactly one class.

- Student must exist first: Parents can only be added if the student already exists in the system.
- NIC required: All teachers, managers, and parents must have a National Identity Card number.
- Subject assignment: Each teacher has one primary subject they teach.
- The system does not include an automated notification system (such as email, SMS, or push notifications) in this phase. In future deployment stages, email or SMS-based notifications can be added once the project moves to a scalable, funded environment.
- One-way communication: Announcements are broadcast only (no replies/comments).
- No multi-school support: System assumes single school deployment.
- Class options: Configurable class sections (e.g., A, B) with extensibility for future expansion.
- A parent and student have a strict one-to-one relationship (one parent per student).
- The system is currently deployed and tested on the demo environment.

Chapter 4

System Architecture

The IskolE system follows the **Model-View-Controller (MVC)** architectural pattern, which separates concerns into three distinct layers to promote modularity, maintainability, and scalability.

4.1 Components & Responsibilities

4.1.1 Model (M)

Role: Handles data and business logic.

Responsibilities:

- Interacts with the database (e.g., fetch, insert, update, delete records).
- Defines data structures (like Students, Teachers, Attendance, etc.).
- Implements application logic (e.g., calculating averages, rankings, attendance percentages).
- Notifies the Controller when data changes.

4.1.2 View (V)

Role: Manages the user interface (UI) and presentation layer.

Responsibilities:

- Displays data from the Model to the user in a readable format (HTML, CSS).
- Provides forms and interfaces for user input.
- Contains no business logic — only handles what the user sees.

4.1.3 Controller (C)

Role: Acts as a bridge between the Model and the View.

Responsibilities:

- Receives input from the user (e.g., form submission, button click).
- Processes the input and requests the appropriate data from the Model.
- Selects which View to display based on the result.
- Controls the overall flow of data and user navigation.

4.2 Key Modules & Interactions

Table 4.1: Key System Modules and Their Interactions

Module	Main Interactions / Purpose
Authentication Module	Handles user login, logout, session management, and password reset functionality. Interacts with User Model and Role Controller.
Student Management Module	Stores student profiles, academic data, and attendance records. Interacts with Teacher and Management modules.
Teacher Management Module	Manages teacher profiles, schedules, and subject assignments. Communicates with Admin and Class modules.
Attendance Module	Takes input (manual or simulated fingerprint), validates with Student Model, and updates attendance tables.
Marks & Performance Module	Allows teachers to enter marks; system calculates averages and ranks. Used by Students, Parents, and Management modules.
Announcement Module	Enables admin, teachers or management to post announcements; students and parents view them via dashboard.

Table 4.1 – continued from previous page

Module	**Main Interactions / Purpose**
Timetable & Calendar Module	Displays schedules and events; data managed by Management panel.
Behaviour & Remarks Module	Teachers record student behaviour; visible to parents and management.
Admin Module	Handles role creation, CRUD operations, and overall system settings such as class, grade, subjects, etc.
Grade & Class Management Module	Create and delete grades and classes. Managed by Administrator.
Subject Management Module	Create and delete subjects. Handled by R. S. R. G. A. A. Ananda.
Assign Class Teacher Module	Assign teachers to their respective classes. Handled by Seniru D S Senaweera.

Chapter 5

System Design

This chapter presents the key design diagrams of the IskolE system, illustrating the structural and behavioural aspects of the software architecture.

5.1 Use Case Diagram

The use case diagrams illustrate the interactions between the system's primary actors (Administrator, Management Panel, Teacher, Student, and Parent) and the functional modules of the system.

5.2 ER Diagram

The Entity-Relationship diagram depicts the database schema, showing all entities, their attributes, and the relationships between them. The system is centred on the `USER` entity, with specialisations for Student, Teacher, and Parent roles.

5.3 Class Diagram

The class diagram presents the MVC-based class structure of the system, showing the relationships between Controllers, Models, and Entity classes.

5.4 System Architecture Diagram

The system architecture diagram provides a high-level overview of the MVC architecture, illustrating how the front-end, back-end, and database layers interact within the system.

Diagrams/Use Case/Administrator.png

Figure 5.1: Administrator Use Case Diagram

Diagrams/Use Case/Management Panel.jpg

Figure 5.2: Management Panel Use Case Diagram

Diagrams/Use Case/Parent UseCase.png

Figure 5.3: Parent Use Case Diagram

Diagrams/Use Case/Student.jpg

Figure 5.4: Student Use Case Diagram

Diagrams/Use Case/Teacher.jpg

Figure 5.5: Teacher Use Case Diagram

Diagrams/ER/iskole_er_diagram.png

Figure 5.6: ER Diagram

Diagrams/class_diagram/class_diagram-2.png

Figure 5.7: Class Diagram

5.5 Activity Diagrams

The activity diagrams depict the flow of operations and interactions for various core functionalities and user roles within the system.

Diagrams/Activity/Admin and workflow.png

Figure 5.8: Admin Workflow

Diagrams/Activity/Manager workFlow.png

Figure 5.9: Manager Workflow

Diagrams/Activity/Parent workflow.png

Figure 5.10: Parent Workflow

Diagrams/Activity/Student workflow.png

Figure 5.11: Student Workflow

Diagrams/Activity/Teacher Workflow.png

Figure 5.12: Teacher Workflow

Diagrams/Activity/User Login and password reset.png

Figure 5.13: User Login and Password Reset Workflow

Chapter 6

Completeness of the Project

6.1 Functionalities Completed

All functionalities defined in the project proposal and interim report have been **100% completed**. The system consists of **16 CRUD-based functional modules**, distributed among four group members as follows:

6.1.1 R K K Jinendra – Module Owner

1. Management Staff Records (CRUD)
2. Teacher Records (CRUD)
3. Student Records (CRUD)
4. Parent/Guardian Records (CRUD)
5. Student Absence Reason Management (CRUD)

6.1.2 Seniru D S Senaweera – Module Owner

1. Announcements (CRUD)
2. Learning Materials Management (Upload/Update/Delete/Download)
3. Teacher Attendance Records (CRUD)
4. Student Attendance Records (CRUD)

6.1.3 S. K. Thasindu Ramsitha – Module Owner

1. Marks / Exam Results Management (Create/Read/Update)

2. Student Timetables (CRUD + publish/unpublish)
3. Teacher Timetables (CRUD + publish/unpublish)
4. Relief / Teacher Substitution Management (CRUD)

6.1.4 R. S. R. G. A. A. Ananda – Module Owner

1. Classes Management (CRUD)
2. Exam Timetables (Upload/Update/Delete + image handling)
3. Teacher Leave Requests (CRUD + approval workflow)
4. Student Behaviour / Remarks (CRUD)
5. Subjects Management (CRUD)

6.2 Functionalities Yet to Complete

There are **no incomplete functionalities**. All planned modules have been successfully implemented and tested.

Chapter 7

Individual Contribution Analysis

All team members contributed equally (25% each) in the following areas:

- System design
- Database design
- Frontend development
- Backend development
- Testing
- Integration

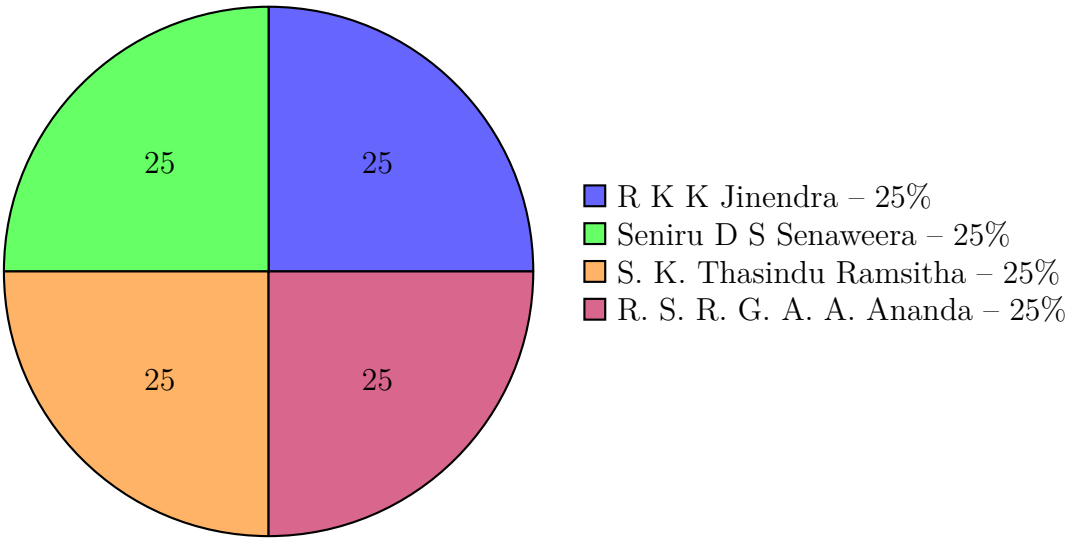

Figure 7.1: Individual Contribution of Team Members

Table 7.1: Individual Contribution Summary

Member	**Contribution (%)**	**Primary Modules**
R K K Jinendra	25%	Staff, Teacher, Student, Parent Records; Absence Reasons
Seniru D S Senaweera	25%	Announcements, Learning Materials, Attendance Records
S. K. Thasindu Ramsitha	25%	Marks/Results, Timetables, Relief Management
R. S. R. G. A. A. Ananda	25%	Classes, Exam Timetables, Leave Requests, Behaviour, Subjects

Chapter 8

Testing

The system testing was conducted using Google Sheets-based test case documentation. The testing process covered multiple levels to ensure system reliability and correctness.

8.1 Testing Types

The following testing methodologies were employed:

1. **Unit Testing:** Individual functions and methods within each module were tested in isolation to verify correctness of business logic and data processing.
2. **Functional Testing:** End-to-end testing of each functional module to ensure that all features operate as specified in the requirements.
3. **CRUD Operation Testing:** Comprehensive testing of all Create, Read, Update, and Delete operations across all 16 functional modules.
4. **Integration Testing:** Testing the interactions between interconnected modules to ensure seamless data flow and consistency.
5. **Role-Based Access Control Testing:** Verification that each user role (Administrator, Management Panel, Teacher, Student, Parent) has access only to their permitted features and data.

8.2 Test Case Summary

Table 8.1: Testing Summary

Aspect	**Details**
Total Functional Units	16 modules
Testing Approach	Module-level and integration testing
CRUD Operations	All operations validated for correctness
Role-Based Access Control	Verified for all 5 user roles
Test Documentation	Google Sheets-based test case tracking
Test Result	All test cases passed successfully

Note: The complete Google Sheets test case document is available at https://docs.google.com/spreadsheets/d/14EDXcn4uQs7-RyXaZ515DG7CmKuiKECqpiORp3X_nPk/edit?usp=sharing.

Chapter 9

Details of Project Supervisor and Co-Supervisor

Proposed Project Supervisor (Academic Staff of UCSC)

Name of the supervisor: Mr. Viraj Welgama

Signature of the supervisor: _____

Date: _____

Proposed Project Co-Supervisor (Assigned by the Course Coordinator)

Name of the co-supervisor: Mr. Janitha Dissanayake

Signature of the co-supervisor: _____

Date: _____

Chapter 10

Declaration

We, as members of the project titled **IskolE**, certify that we will carry out this project according to the guidelines provided by the coordinators and supervisors of the course, as well as that we will not incorporate, without acknowledgment, any material previously submitted for a degree or diploma in any university.

To the best of our knowledge and belief, the project work will not contain any material previously published or written by another person or ourselves except where due reference is made in the text at appropriate places.

Diagrams/member details.png